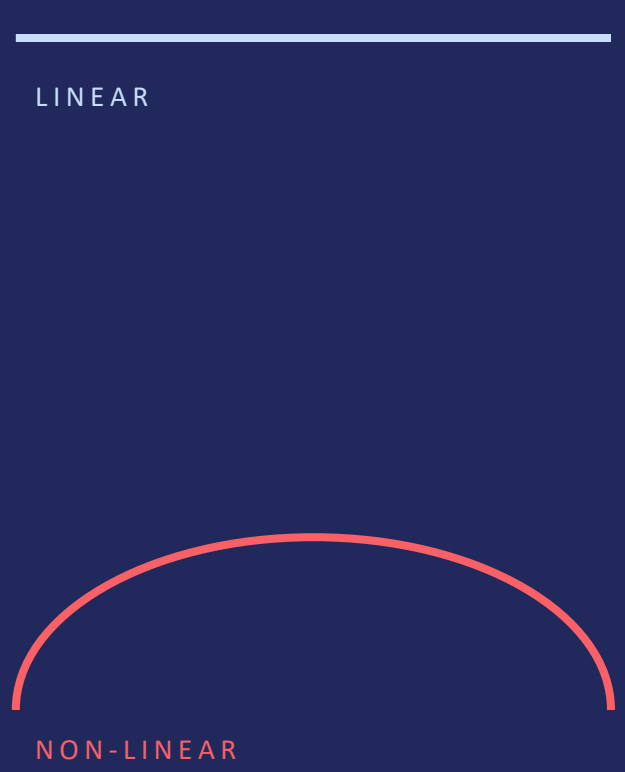
MACHINE LEARNING · UNIT 2

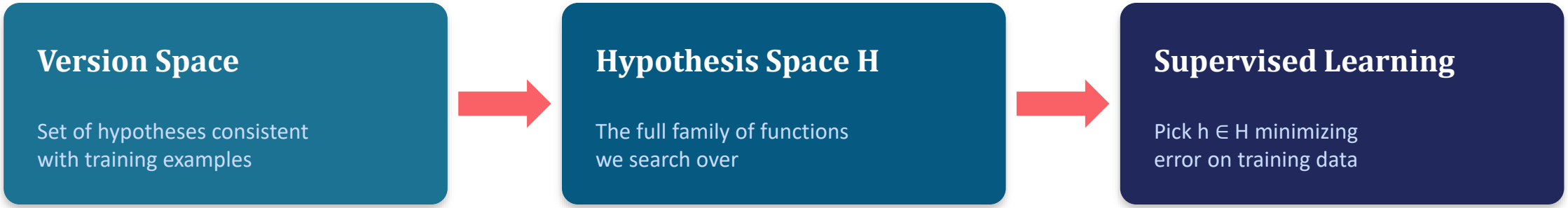
Supervised Learning – I

Linear and Non-Linear Examples

From straight-line hypotheses to curved decision boundaries — and why both matter.



From Version Spaces to Supervised Learning



QUICK RECAP

Every Model Lives in a Hypothesis Space

The hypothesis space is the full set of candidate functions a learning algorithm is allowed to pick from. Choosing a model type — linear, polynomial, tree, network — is choosing that space before we ever see the data.

Small / Simple Space

Straight lines only. Fast, interpretable, but may underfit curved patterns.

Large / Flexible Space

Curves, trees, networks. Powerful, but risks overfitting noise.

PART 1

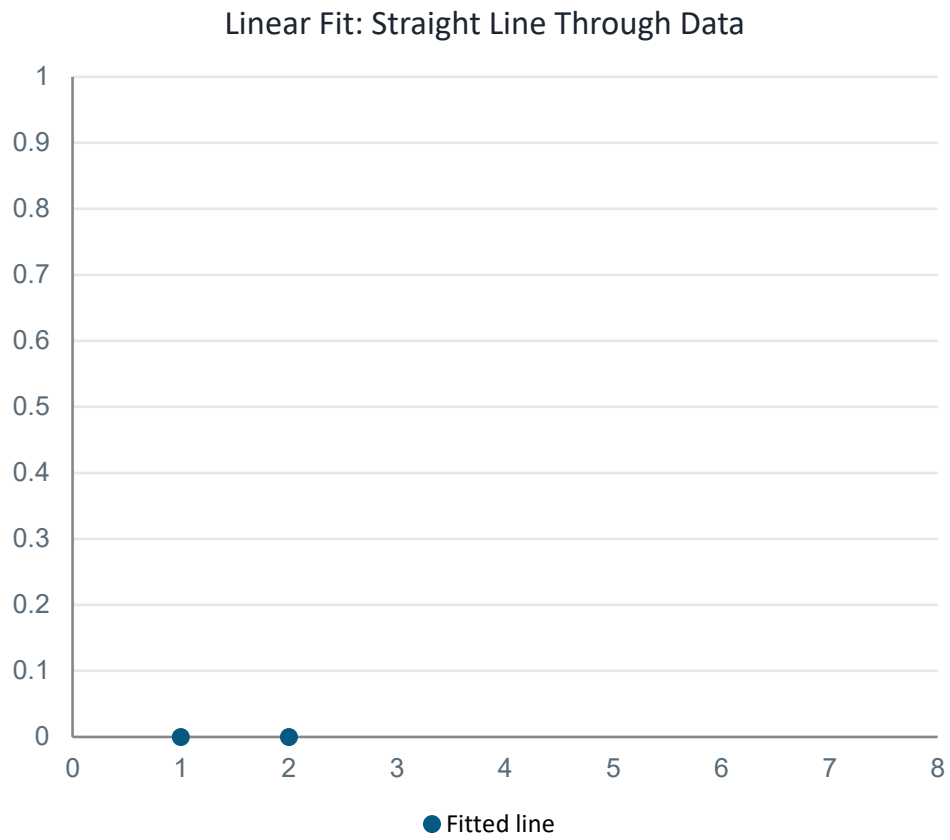
Linear Models

The output is modeled as a weighted sum of the inputs, plus a constant. Geometrically: a straight line, plane, or hyperplane — no bends.

$$y = w_1x_1 + w_2x_2 + \ldots + w_nx_n + b$$

When it works well:

The true relationship is roughly proportional or additive — e.g., more years of experience tends to raise salary in a fairly steady way.



PART 1

Linear Models in Practice

⬡

Linear Regression

Predict house price from square footage: price = m × (sq ft) + b

◉

Logistic Regression

Classify emails as spam / not spam from a weighted sum of word-frequency features

PART 2

Non-Linear Models in Practice

1

Polynomial Regression

Fit a curve $y = ax^2 + bx + c$ — e.g., the trajectory of a thrown ball over time

2

Decision Trees

If-else rule splits carve out non-linear regions: “age < 30 AND income > 50k → approve”

3

Kernel SVM (RBF)

Separates classes with a curved boundary when data isn’t linearly separable

4

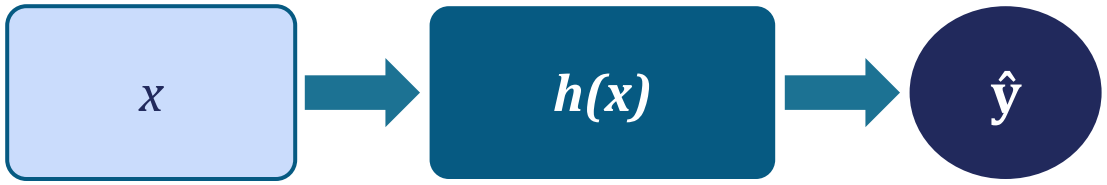
Neural Networks

Non-linear activations (ReLU, sigmoid) model highly complex patterns, e.g. image recognition

What Is Supervised Learning?

Learning a mapping from inputs to outputs using labeled examples.

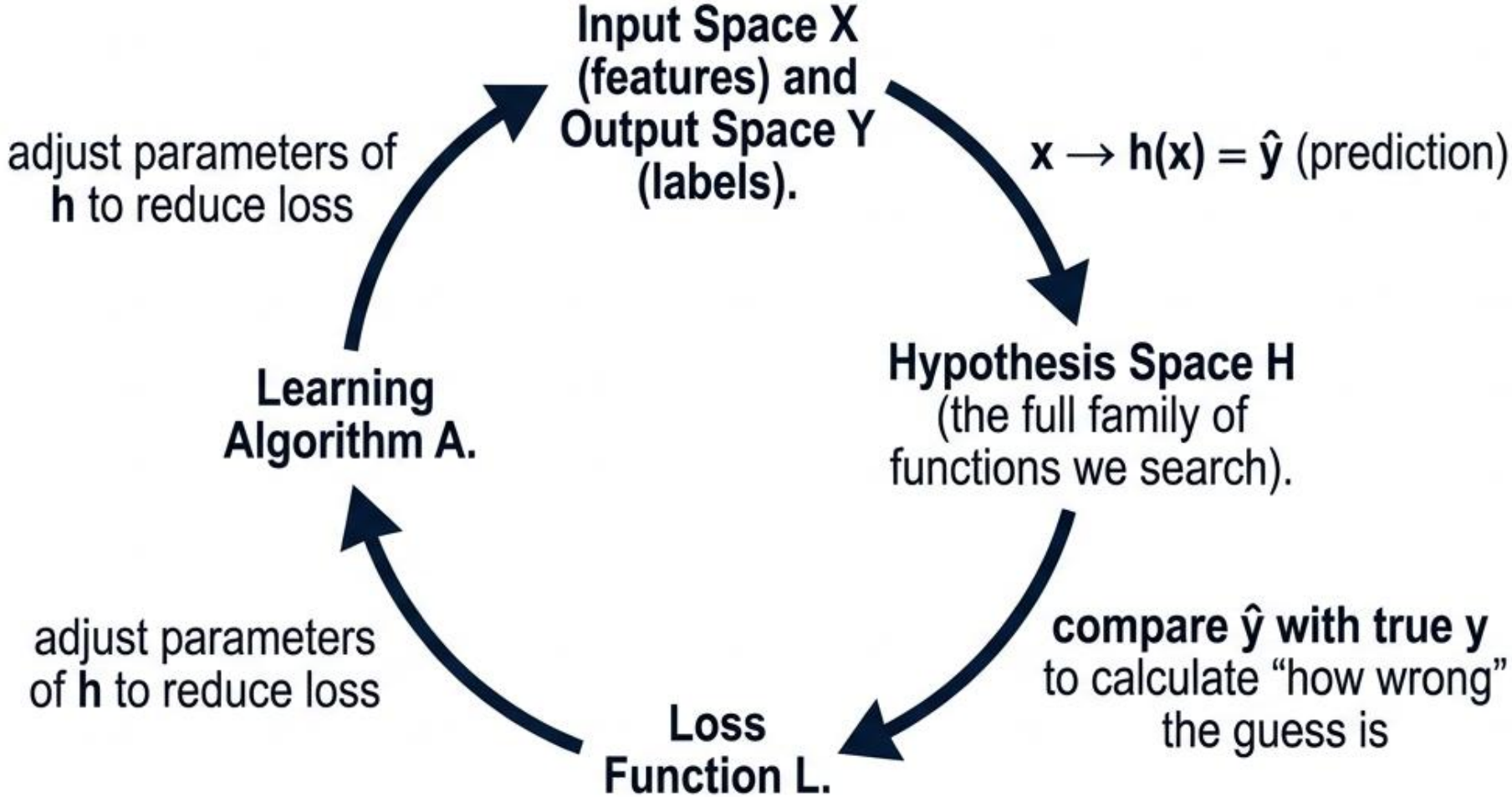
- **X** Input space (features)
- **Y** Output space (labels)
- **H** Hypothesis space we search
- **L** Loss function — how wrong h(x) is
- **A** Learning algorithm — how we search H



compare with true y → Loss

adjust parameters of h to reduce loss

The Anatomy of Supervised Learning is a Continuous Optimization Loop



Supervised learning isn't just about finding any consistent hypothesis—it is about deliberately navigating H to minimize error on training data.

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Types of Regression Models

- Based on the type of functions used to represent the relationship between the dependent or output variable and independent variables, the regression models are categorized into four types. The regression models are,

1. Simple Linear Regression
2. Multiple Regression
3. Polynomial Regression
4. Logistic Regression

1. Simple Linear Regression

- Assume that there is only one independent variable x. If the relationship between x (independent variable) and y (dependent or output variable) is modeled by the relation,

$$y = a + bx$$

- then the regression model is called a linear regression model.

2. Multiple Regression

- Assume that there are multiple independent variables *say* x_1, x_2, ...x_n. If the relationship between independent variables x and dependent or output variable y is modeled by the relation,

$$y=a_0+a_1*x_1+a_2*x_2+.....+a_n*x_n$$

- then the regression model is called a multiple regression model.

3. Polynomial regression

- Assume that there is only one independent variable x. If the relationship between independent variables x and dependent or output variable y is modeled by the relation,

$$y=a_0+a_1*x+a_2*x^2+\ldots\ldots\ldots+a_n*x^n$$

- for some positive integer n>1, then we have a polynomial regression.

4. Logistic Regression

- Logistic regression is used when the dependent variable is binary (0/1, True/False, Yes/No) in nature.

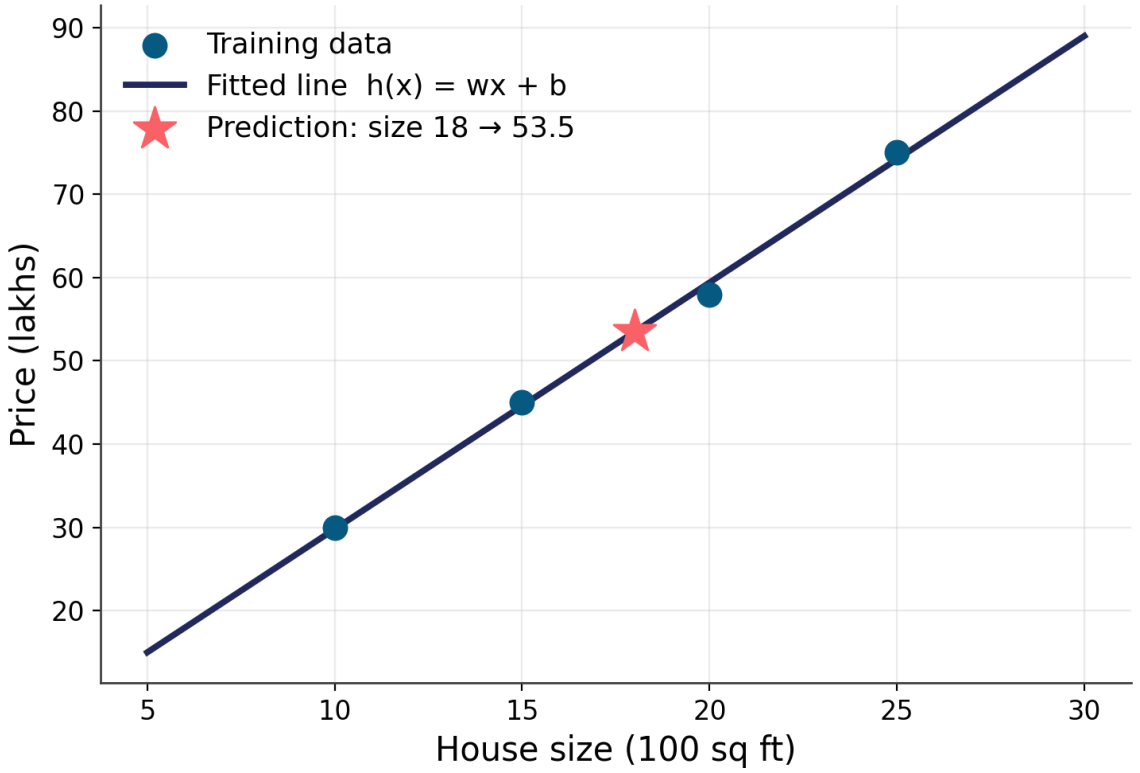
Linear Regression — House Price Example

$$h(x) = w \cdot x + b$$

Fit a straight line through the data by minimizing squared error:

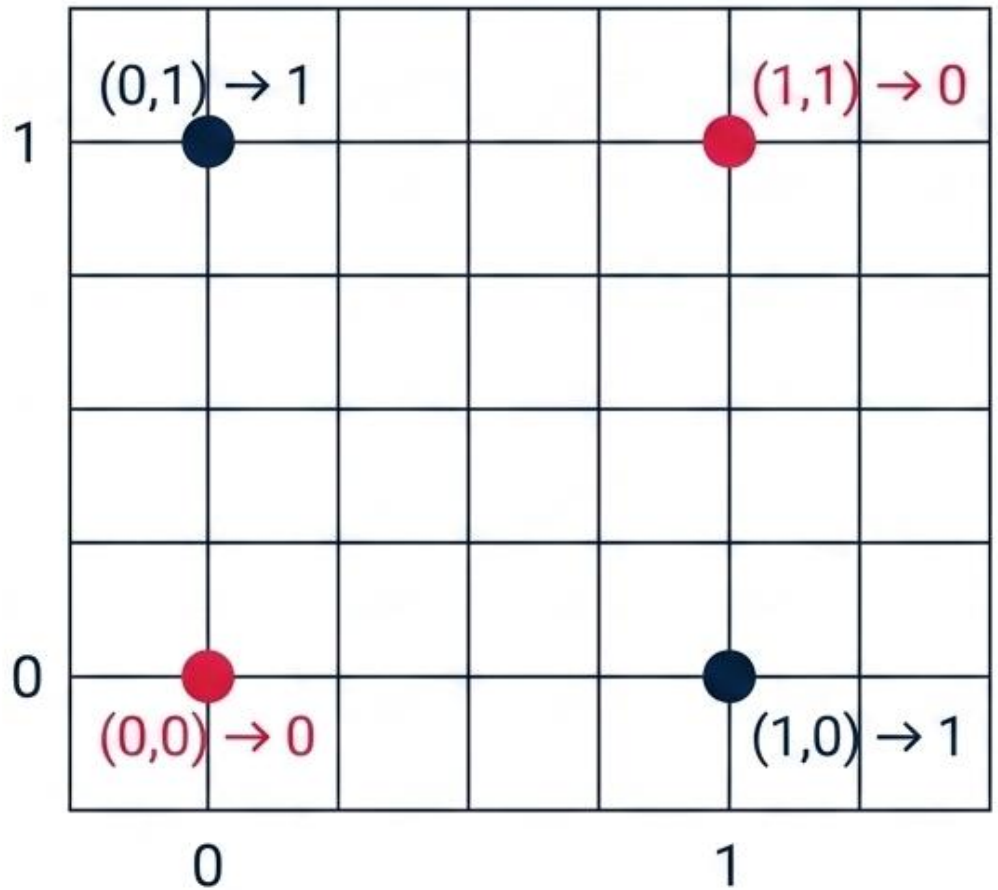
$$J(w,b) = (1/n)\ \Sigma\ (h(x_i) - y_i)^2$$

- 4 data points: house size vs. price
- Least squares gives w ≈ 2.96, b ≈ 0.20



Ask class to predict the price for size 18 BEFORE revealing the fitted line.

The Limit of Rigidity: Why Linear Models Fail the XOR Problem



0, 0 → 0
0, 1 → 1
1, 0 → 1
1, 1 → 0

The Consequence: No single straight line can possibly separate the crimson classes from the ink-navy classes.

The Result: A linear perceptron on XOR gets permanently stuck at 50% accuracy, regardless of training time. This limitation birthed the historical motivation for multi-layer networks and kernel methods.

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Non-Linear Regression — Polynomial Fit

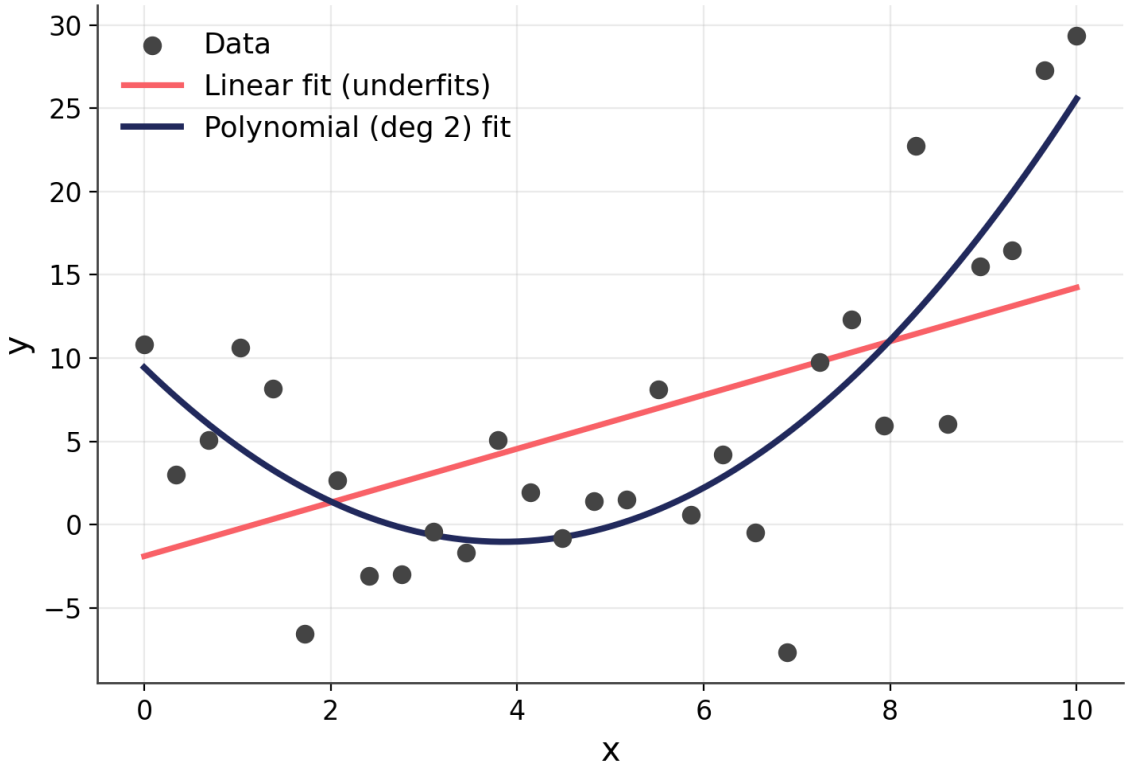
$$h(x) = w_0 + w_1 x + w_2 x^2$$

Curved data — a straight line underfits.

Still linear in the parameters w, but the curve traced in x is non-linear.

Fit quality (R^2, 1.0 = perfect):

- **Linear model:** $R^2 \approx 0.29$



Discussion: could we keep raising the degree forever and always do better? (→ overfitting, future class)

THE TRADE-OFF

Choosing Between Them

Linear Models

Strengths

✓ Simple and fast to train
✓ Easy to interpret weights
✓ Less prone to overfitting

Limitation

✗ Cannot capture curved or complex patterns (underfitting risk)

Non-Linear Models

Strengths

✓ Captures complex, curved relationships
✓ More flexible and powerful

Limitations

✗ Needs more data to train well
✗ Prone to overfitting noise
✗ Often harder to interpret (“black box”)

SYNTHESIS

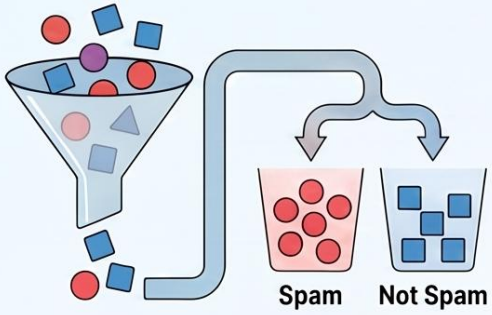
Linear vs Non-Linear Models

Aspect	Linear Models	Non-Linear Models
Hypothesis space	Simple (lines / hyperplanes)	Complex (curves, trees, surfaces)
Interpretability	High	Often lower
Underfitting risk	Higher on complex data	Lower
Overfitting risk	Lower	Higher — needs regularization / data
Computation	Cheap	Often expensive
Example algorithms	Linear/Logistic Regression, Perceptron	Decision Trees, k-NN, Kernel SVM, Neural Nets

Supervised Learning 101: Linear vs. Non-Linear Models

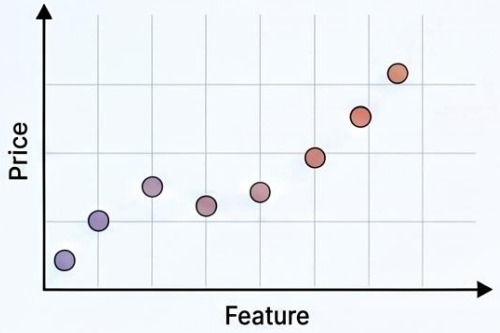
The Two Tasks of Supervised Learning

Classification



Output is discrete/categorical with clear boundaries, such as identifying spam or tumors.

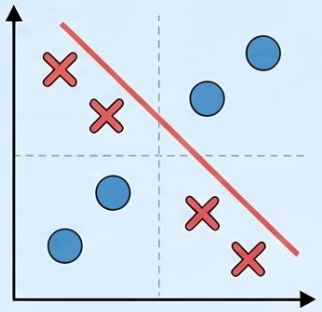
Regression



Output is a continuous, ordered numeric value, such as predicting house prices.

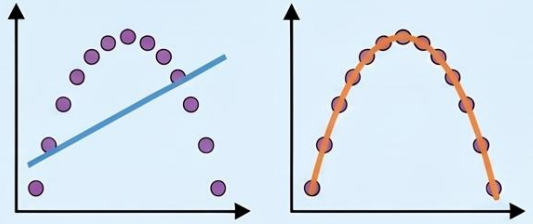
Linear vs. Non-Linear Trade-offs

The Limit of Linear Models



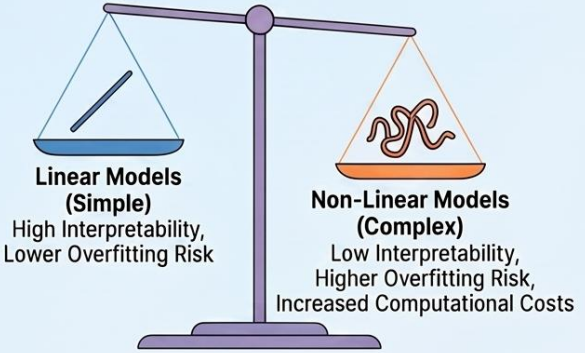
Simple models fail on XOR problems where no straight line can separate classes.

Polynomial Expressive Power



In curved data, a degree-2 polynomial ($R^2 \approx 0.68$) outperforms linear fits ($R^2 \approx 0.29$).

The Complexity Trade-off



Non-linear models reduce underfitting but increase the risk of overfitting and computational costs.

Linear Models		VS	Non-Linear Models
Hypothesis Space	Simple (Lines/Hyperplanes)		Complex (Curves, Surfaces)
Interpretability	HIGH	VS	LOW
Examples	Linear Regression, Perceptron		Decision Trees, k-NN, SVM

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